Red Hat
Process Automation

Software and Services for Business Automation

PV207 BPM

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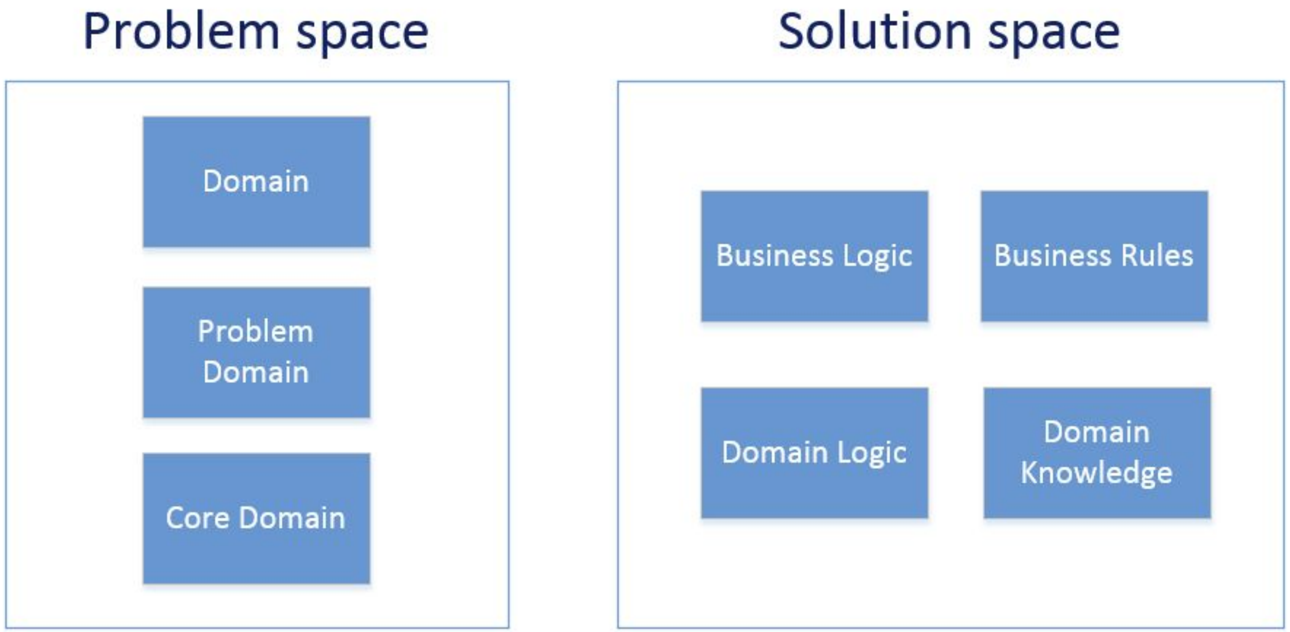
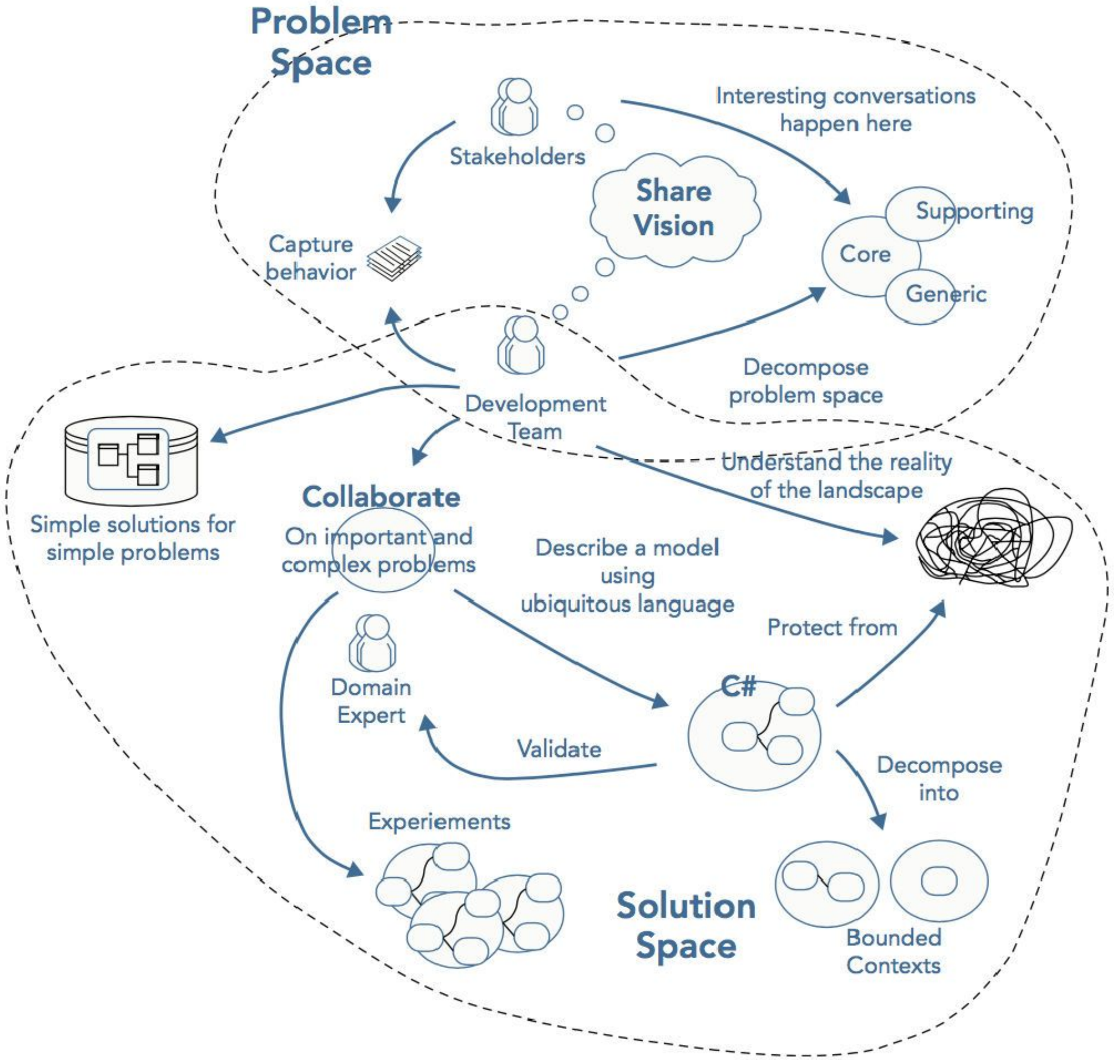
DOMAIN-DRIVEN DESIGN (DDD)

Domain-driven design is a concept that the structure and language of software code should match the business domain.



- Eases communication
- Improves flexibility
- Emphasizes domain over interface
- Requires robust domain expertise
- Encourages iterative practices
- Ill-suited for highly technical projects

DOMAIN-DRIVEN DESIGN (DDD)



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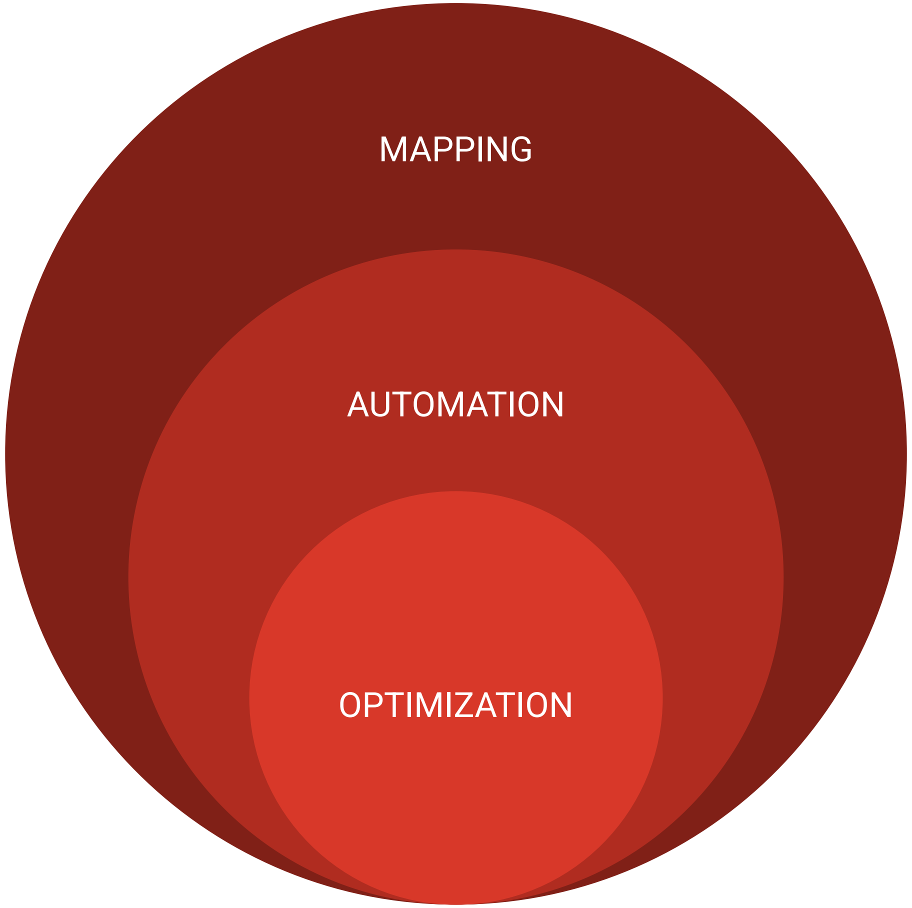
BUSINESS AUTOMATION



MARKET

Low Code Platforms

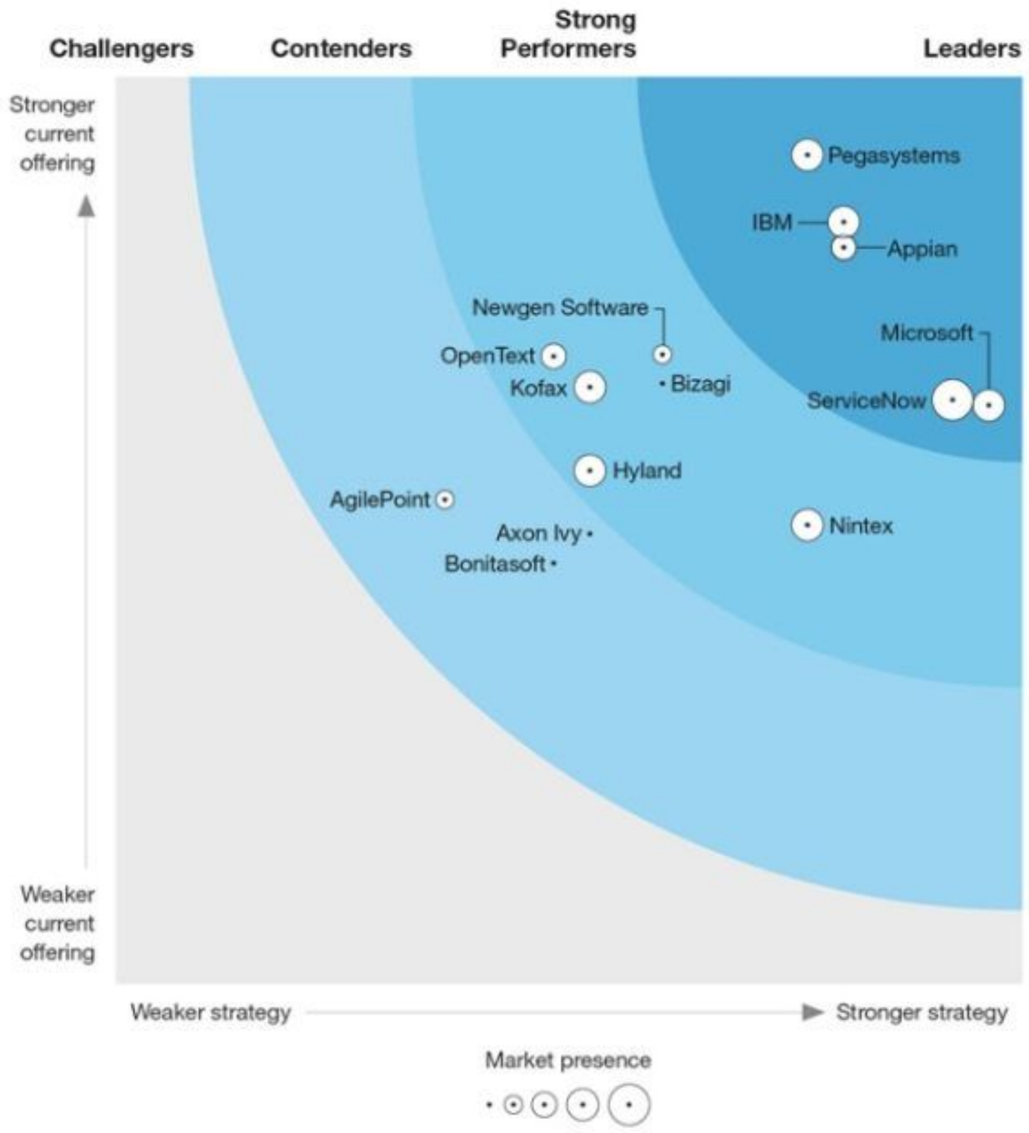
DPA Platforms

Cloud Services

Communities

(Object Management Group, Cloud Native Computing Foundation, Kubernetes, Knative, KIE, Kogito)

DPA SOFTWARE FROM FORRESTER ANALYST REPORT '21



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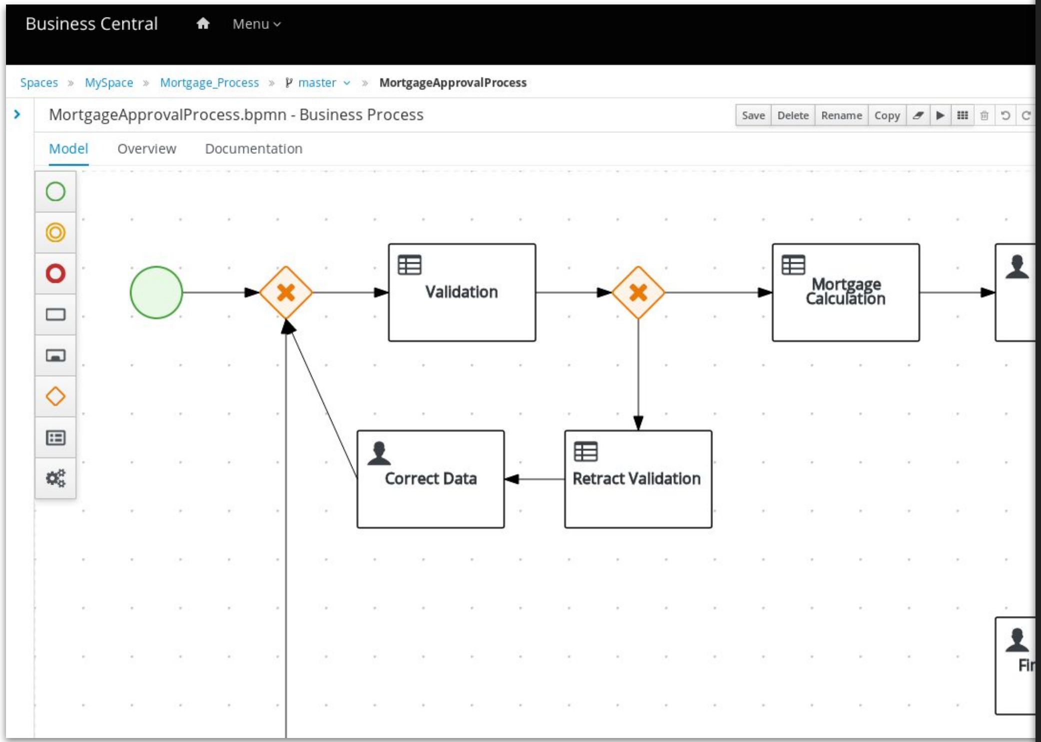
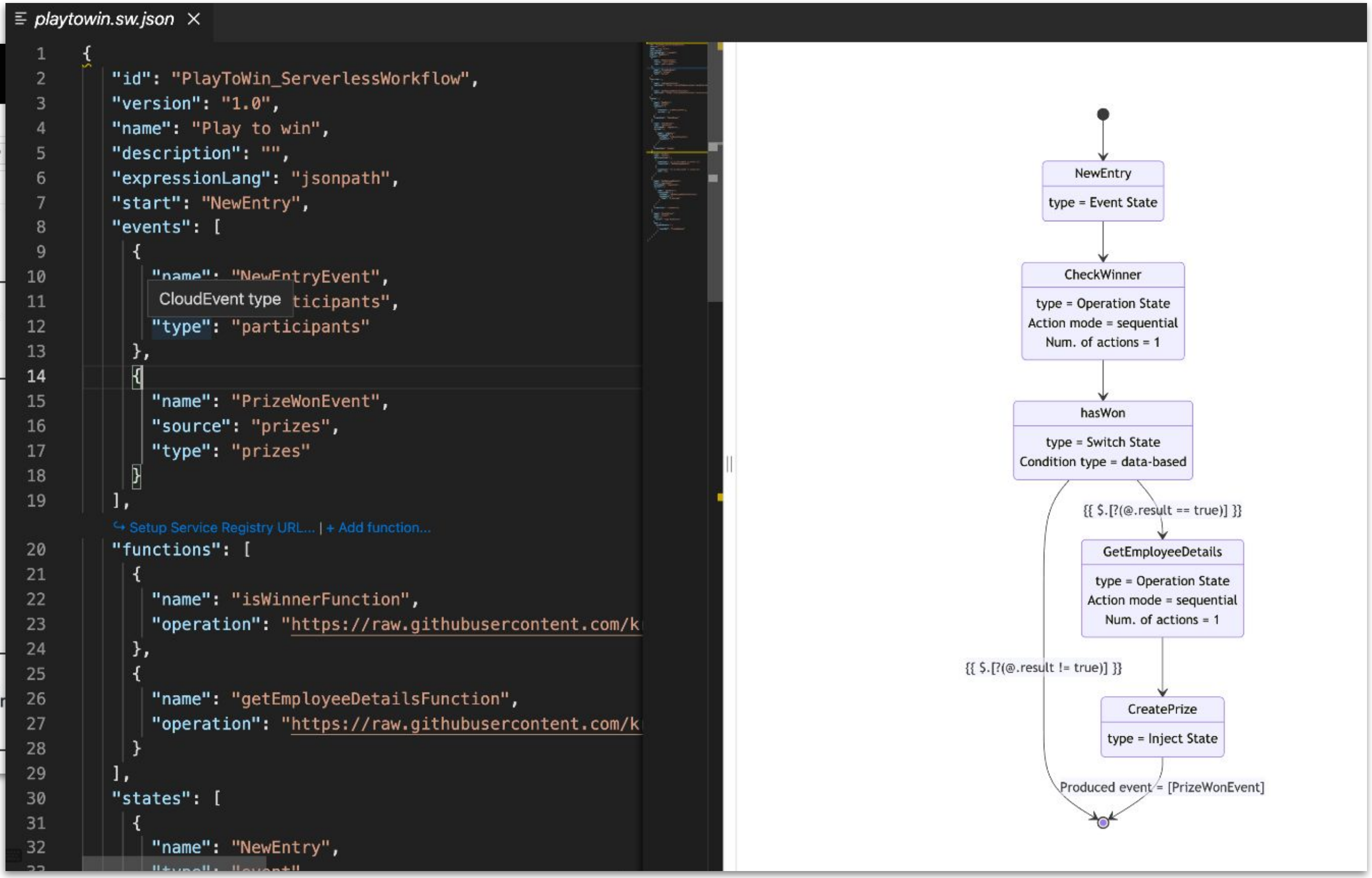
AUTOMATION OF WORKFLOWS

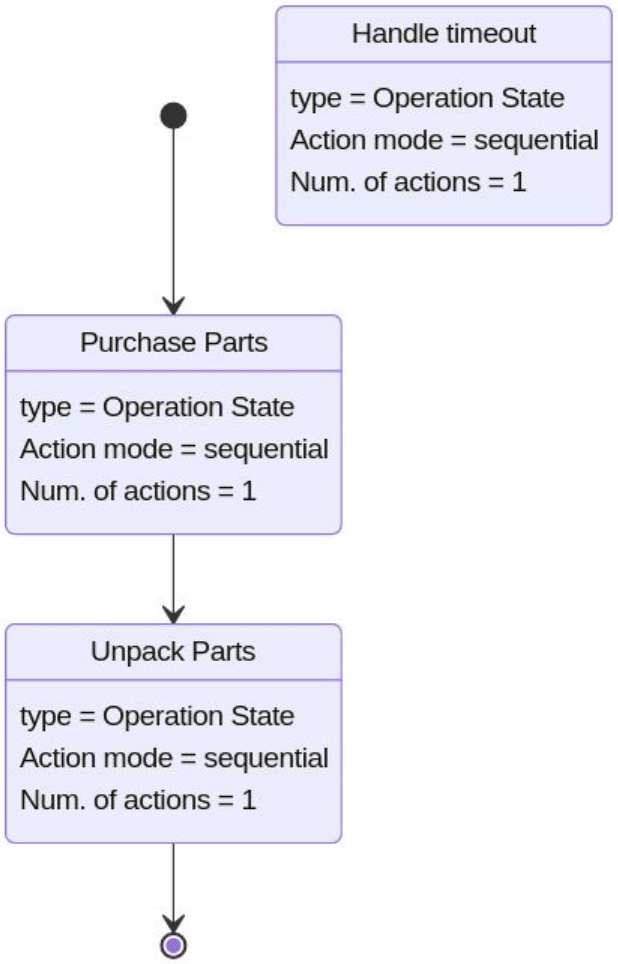




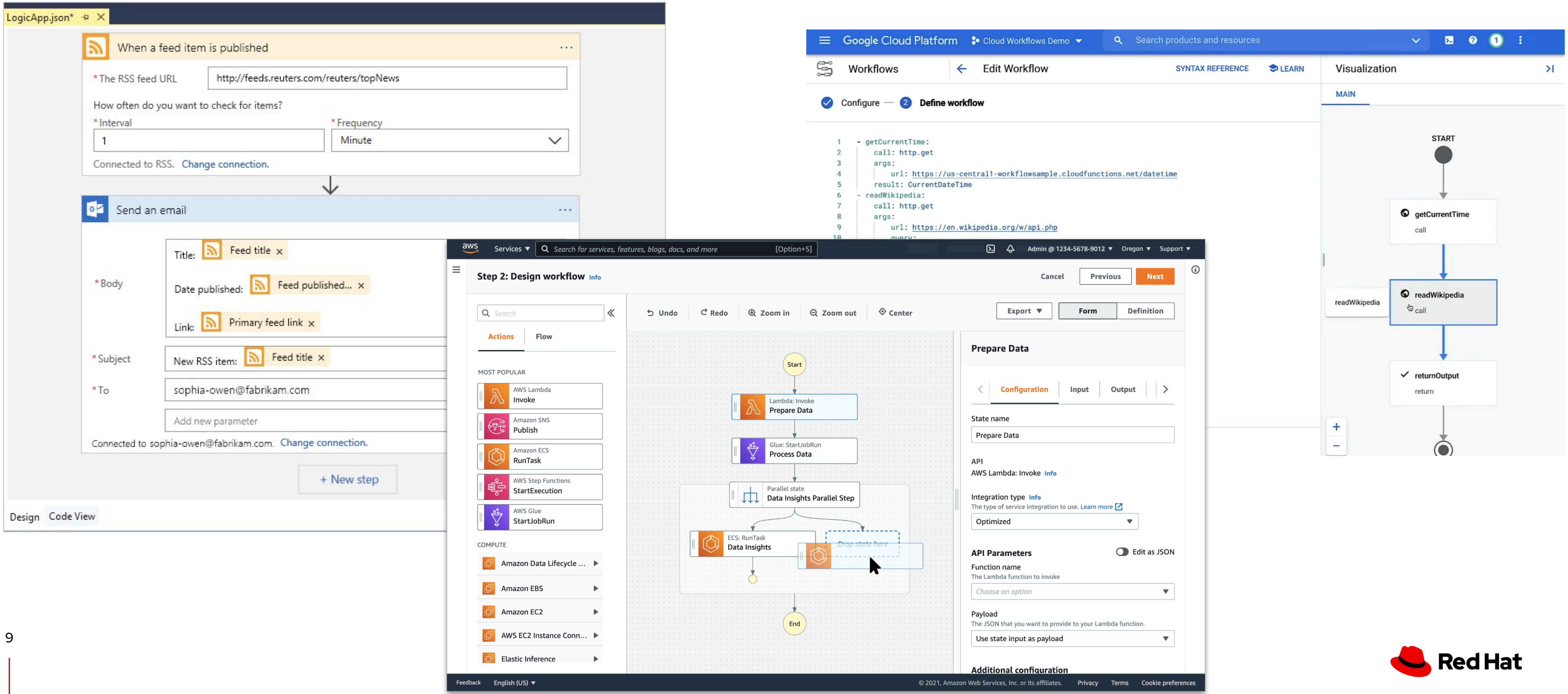
SERVERLESS WORKFLOW

- Similar specification to BPMN but aimed more at developers
- More lightweight, no human tasks, targeted at machine interaction
- Workflow is “modelled” directly by writing the code (JSON/YAML), currently doesn’t define graphical notation
- Higher level than BPM, focuses more on orchestration of services or infrastructure in the cloud, self-healing infrastructure
- Background service, end users don’t interact directly with it (as with BPM in loan approval, order processing…)
- https://serverlessworkflow.io/
- BPMN vs. SWF

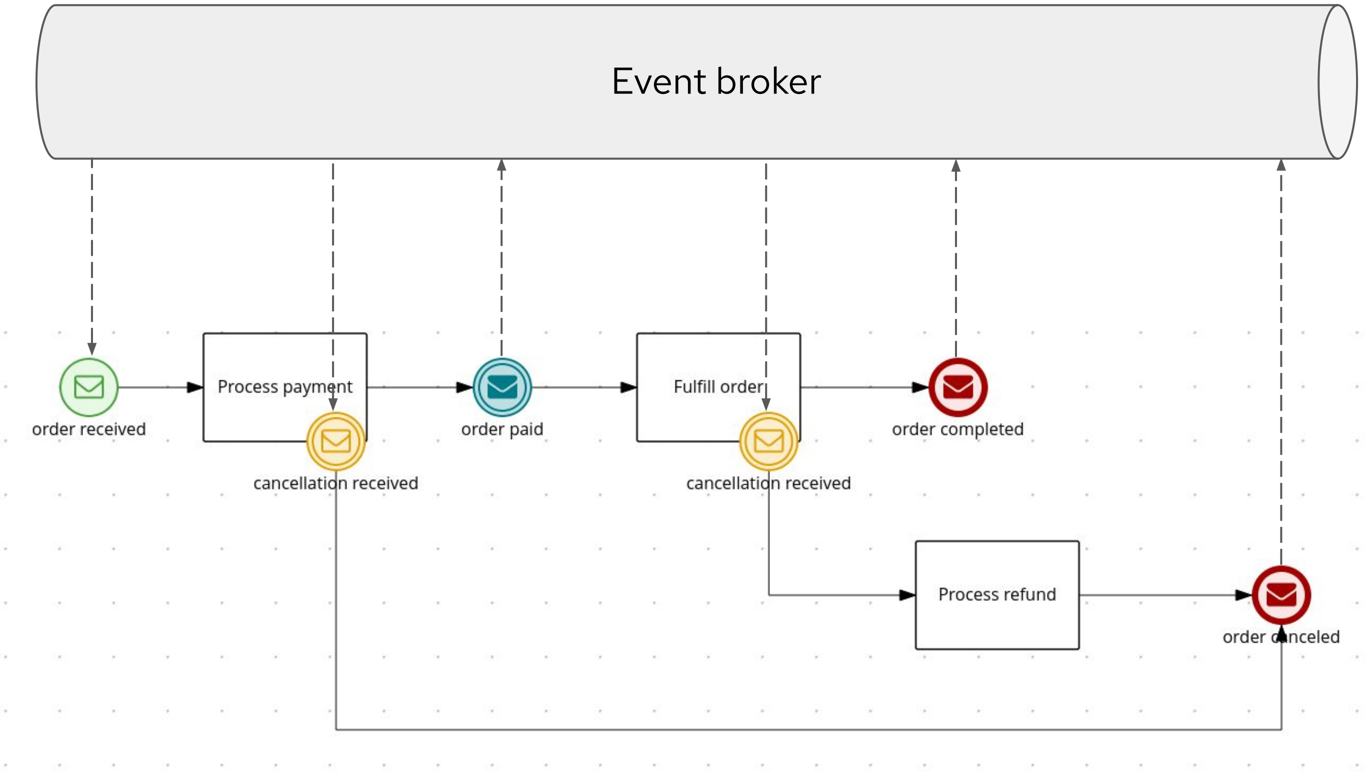
```yaml
id: executiontimeout
name: Execution Timeout Workflow
version: '1.0.0'
specVersion: '0.8'
start: Purchase Parts
timeouts:
  workflowExecTimeout:
    duration: PT7D
    interrupt: true
    runBefore: Handle timeout
states:
- name: Purchase Parts
  type: operation
  actions:
  - functionRef: purchasePartsFunction
  transition: Unpack Parts
- name: Unpack Parts
  type: operation
  actions:
  - functionRef: unpackPartsFunction
  end: true
- name: Handle timeout
  type: operation
  actions:
  - functionRef: handleTimeoutFunction
functions:
- name: purchasePartsFunction
  operation: file://myservice.json#purchase
- name: unpackPartsFunction
  operation: file://myservice.json#unpack
- name: handleTimeoutFunction
  operation: file://myservice.json#handle
```



AUTOMATION OF WORKFLOWS

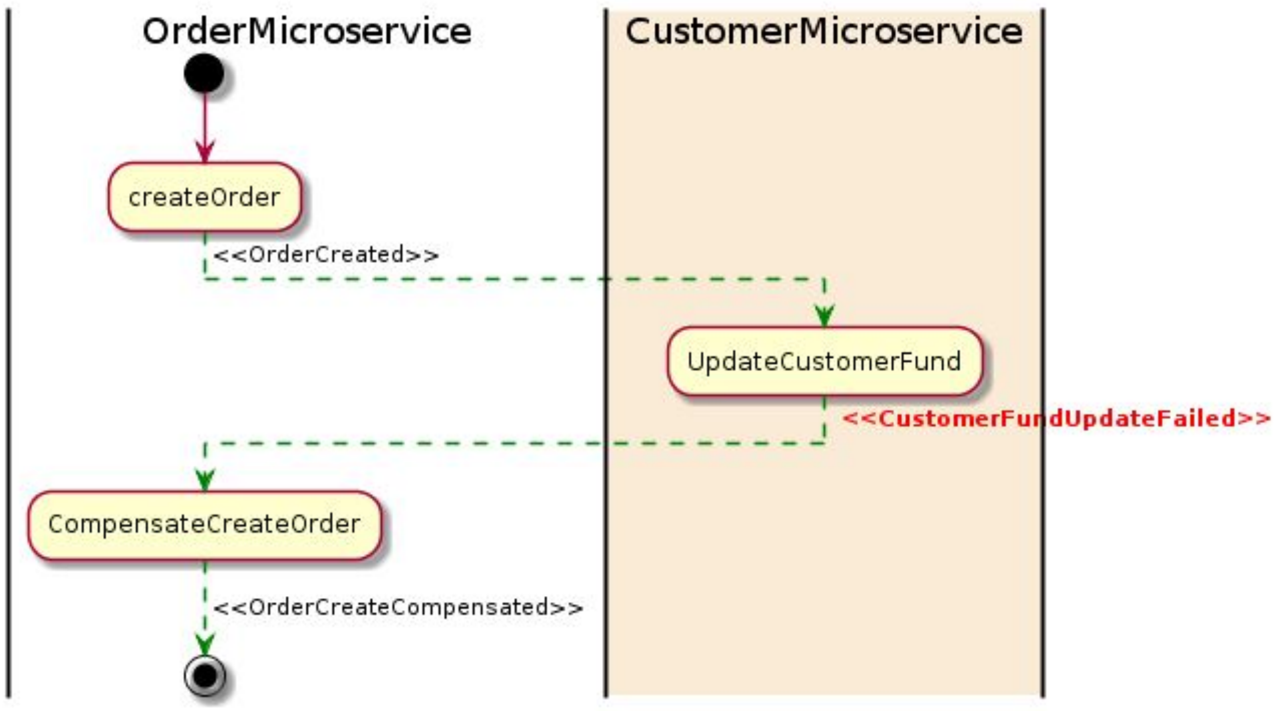


AUTOMATION OF EVENT-DRIVEN WORKFLOWS



AUTOMATION OF LONG-LIVED TRANSACTIONS

Saga pattern for microservice architectures



If any microservice fails to complete its local transaction, the other microservices will run **compensation** transactions to rollback the changes.

<u>Advantages</u>

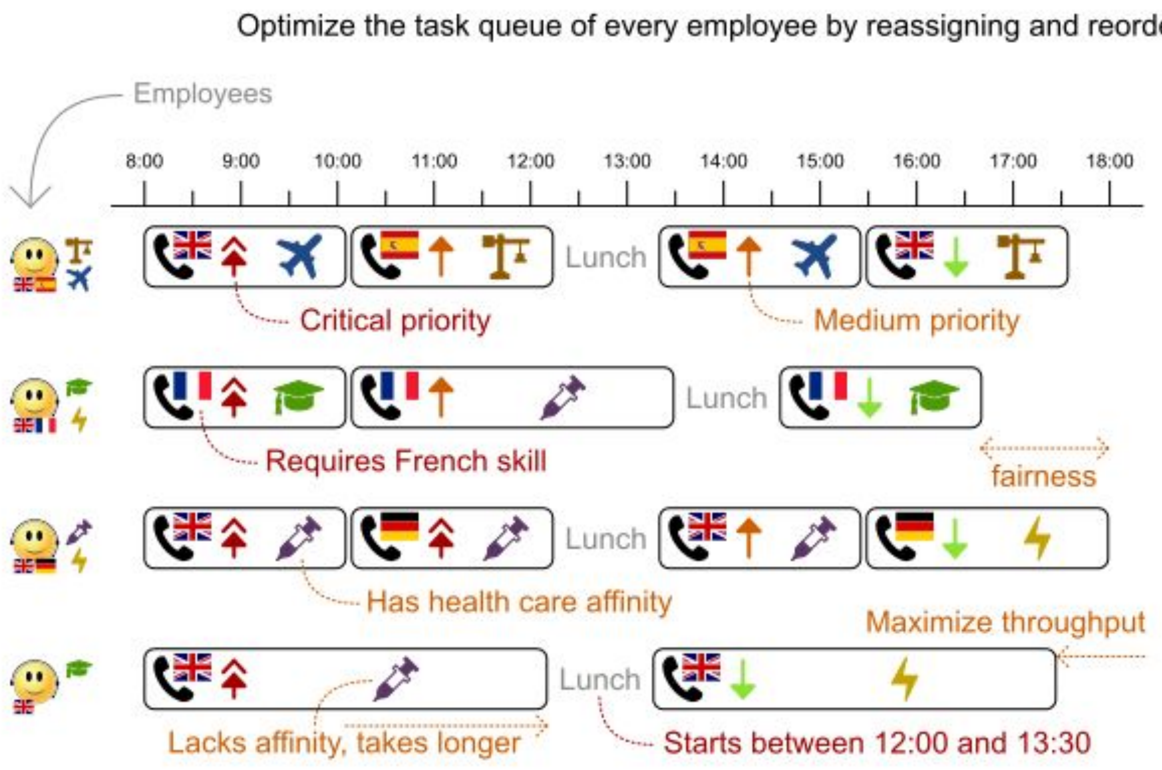
- support for long-lived transactions
- other microservices are not blocked if a microservice is running for a long time
- there is no lock on any object

<u>Disadvantages</u>

- difficult to debug
- difficult to maintain if the system gets complex
- does not have read isolation

Adding a **process manager** addresses the complexity issue of the Saga pattern when it becomes responsible for listening to events and triggering endpoints.

AUTOMATION OF TASK ASSIGNMENTS



Users

Payroll services

Call centers

Tax auditers

Recruitement interviewing

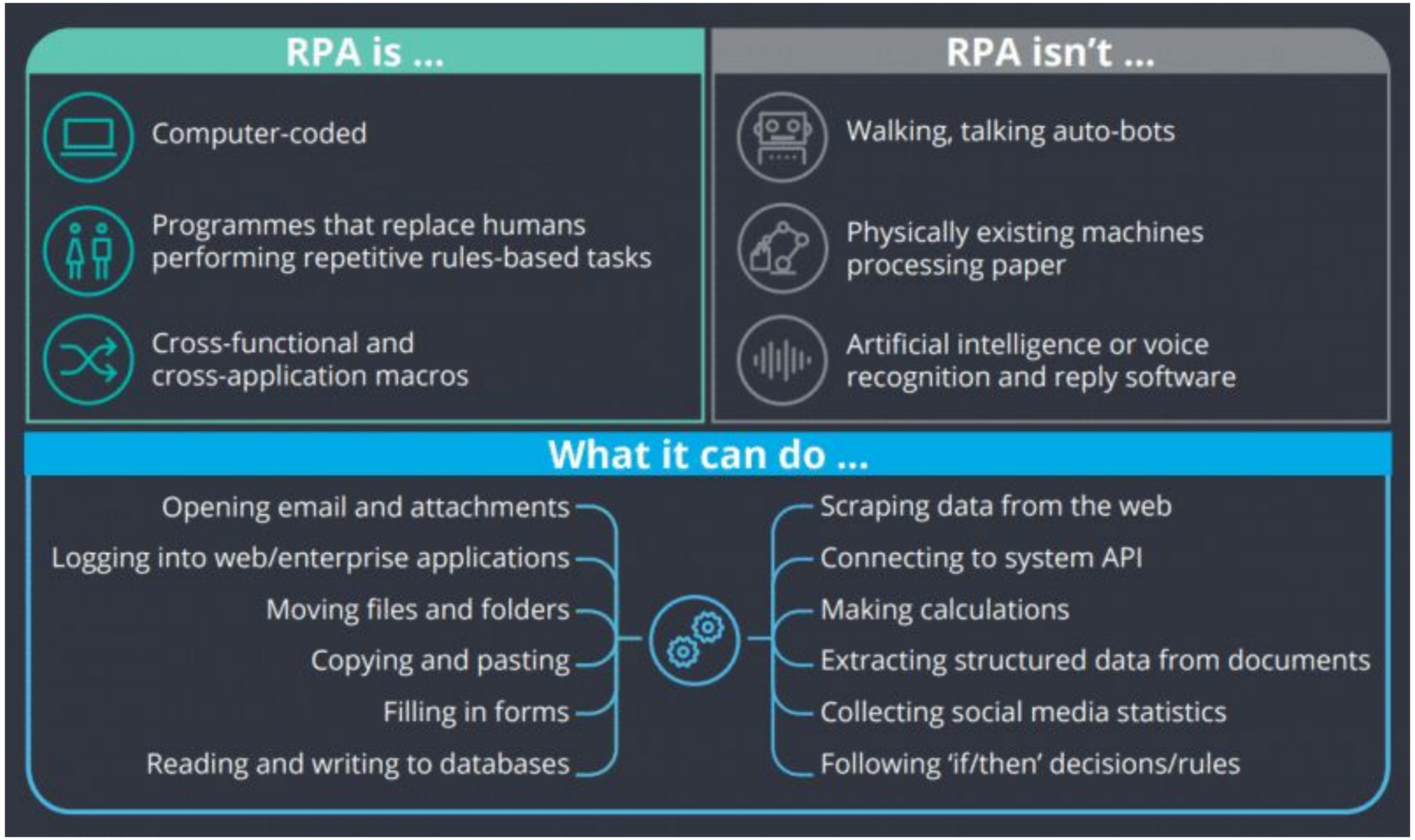
Mortgage approval

[KIELive#45] Optimizing user tasks assignment in business processes with Kogito and OptaPlanner*

* OptaPlanner has been succeeded by timefold

AUTOMATION OF HUMAN ACTIONS INTERACTING WITH SW

Robotic Process Automation (RPA)



AUTOMATION OF BUSINESS RULES

Kogito DRL Rules language

IBM Action rules express business policy statements using a predefined business vocabulary that can be interpreted by a computer.

Policy

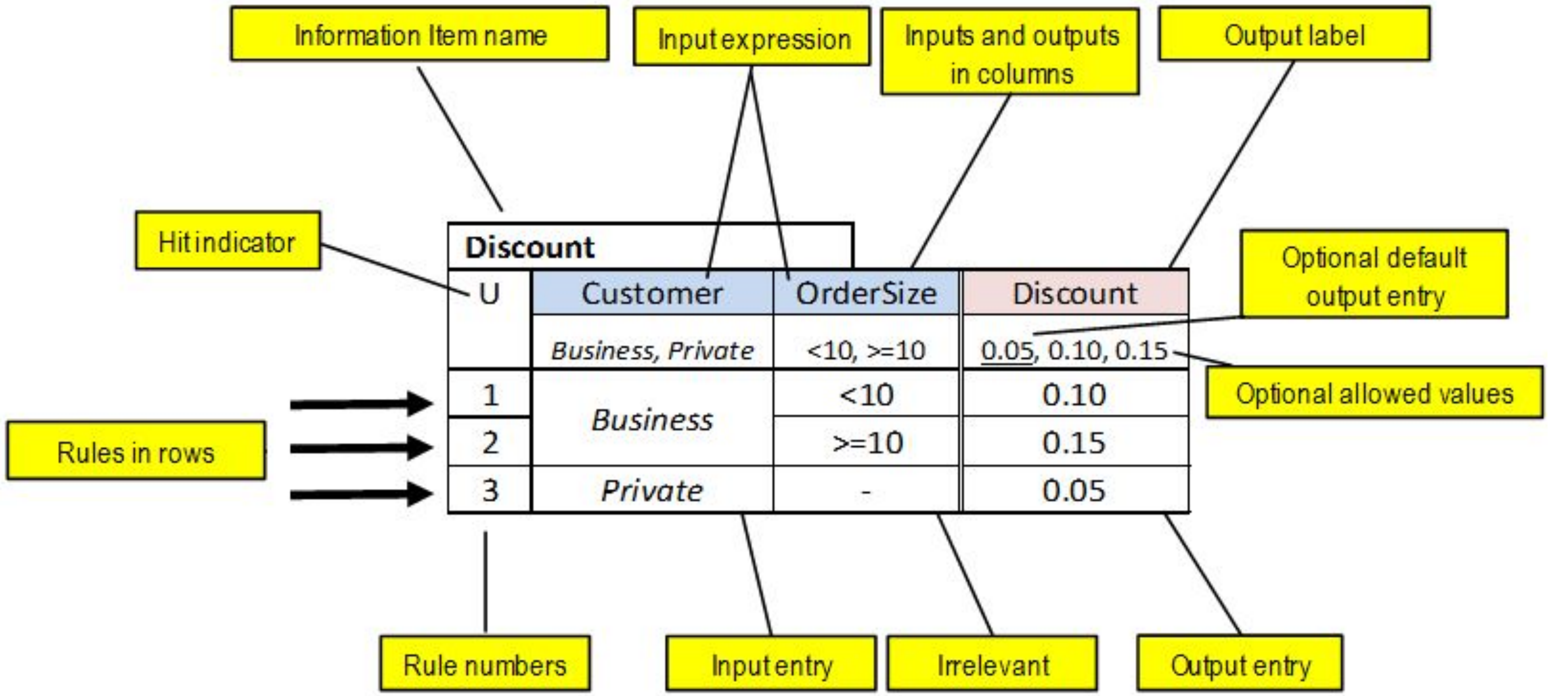
"change customers in the `Gold` category to the `Platinum` category when they spend more than `$1,500` in a single transaction"

Action Rule

If All of the following conditions are true:
- the customer category is `Gold`
- the value of the shopping cart is more than `$ 1,500`

Then Change the customer category to `Platinum`

AUTOMATION OF DECISION TABLES

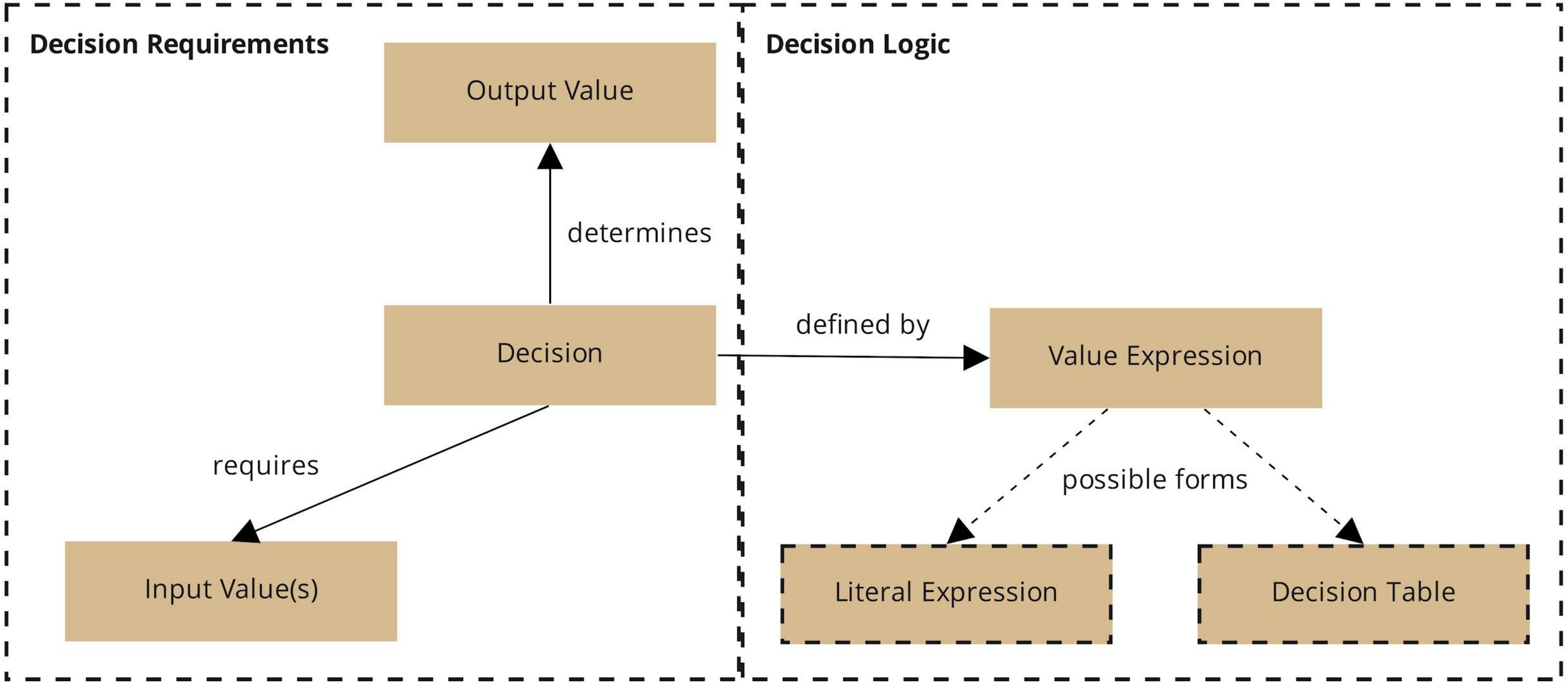


Types of decisions

- Selection (routing)
- Scoring
- Categorizing

The decision doesn't take an action (no side effect), just determines a data value

AUTOMATION OF DECISION MODELS



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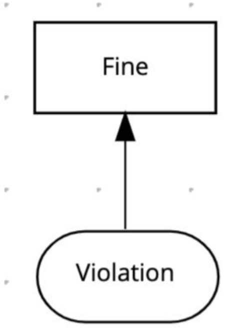
AUTOMATION OF DECISION MODELS

Executing decision logic in DMN models



Fine *(Decision Table)*

U	Violation.Type *(string)*	Violation.Actual Speed - Violation.Speed Limit *(number)*	Fine *(tFine)*		Description
			Amount *(number)*	Points *(number)*	
1	"speed"	[10..30)	500	3	
2	"speed"	>= 30	1000	7	
3	"parking"	-	100	1	
4	"driving under the influence"	-	1000	5	

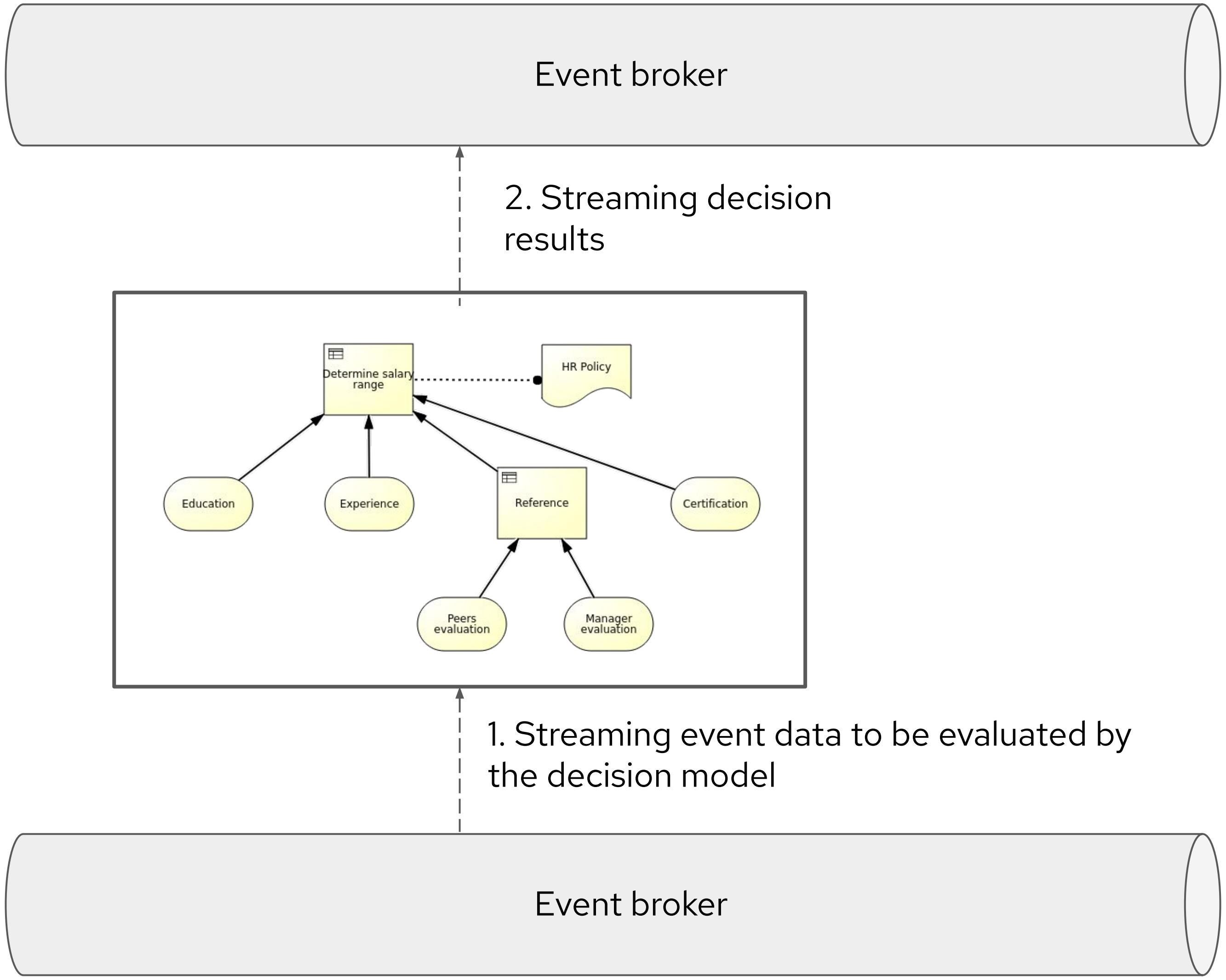
```
/* = Actual input data = */
"Violation": {
  "Type": "speed",
  "Speed Limit": 60,
  "Actual Speed": 100
}
```

```
/* = Expected output data = */
"Fine": {
  "Points": 7,
  "Amount": 1000
}
```

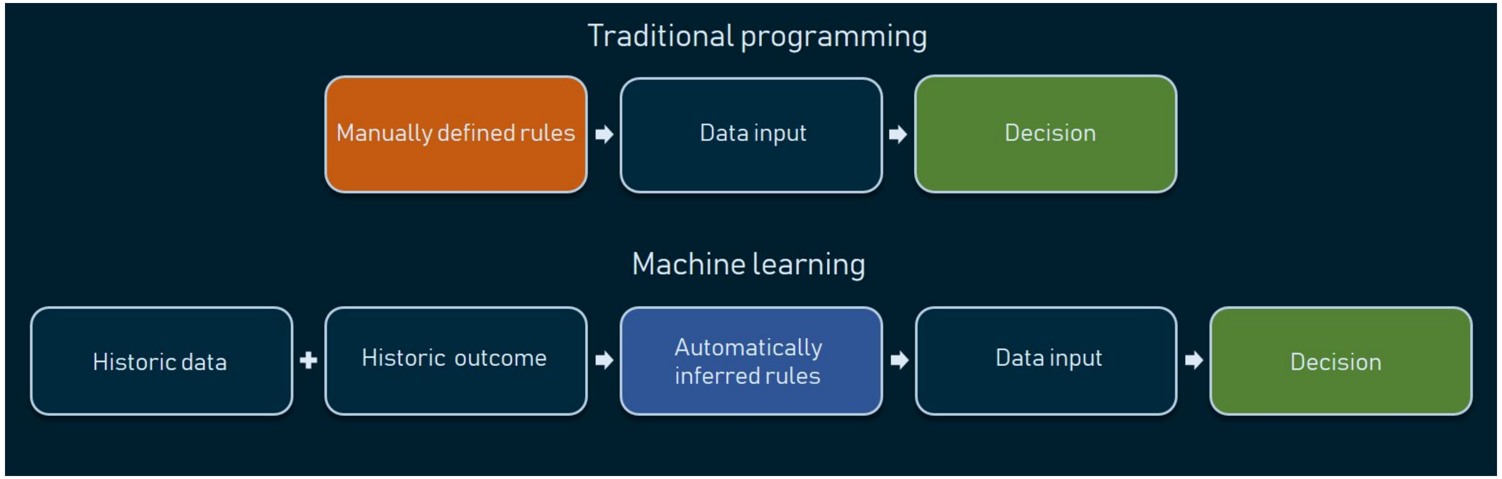
[DMN with Kogito on Quarkus]

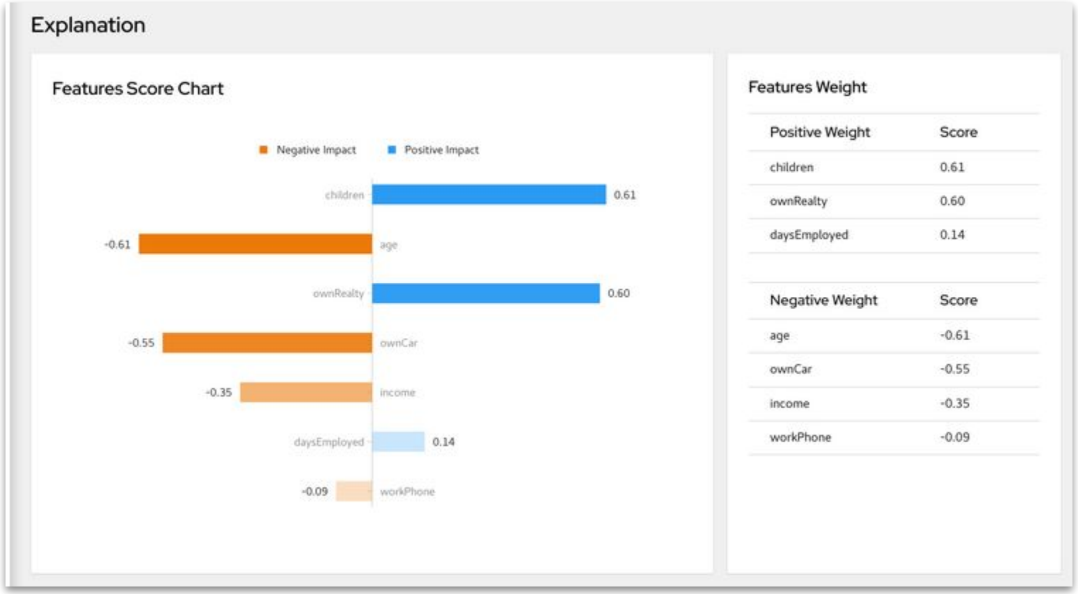
[DMN FEEL Cheatsheet]

AUTOMATION OF DECISION MAKING IN STREAMS



AUTOMATION OF DECISION MAKING WITH AI





[TrustyAI]()

[Can you trust AI? - presentation]()

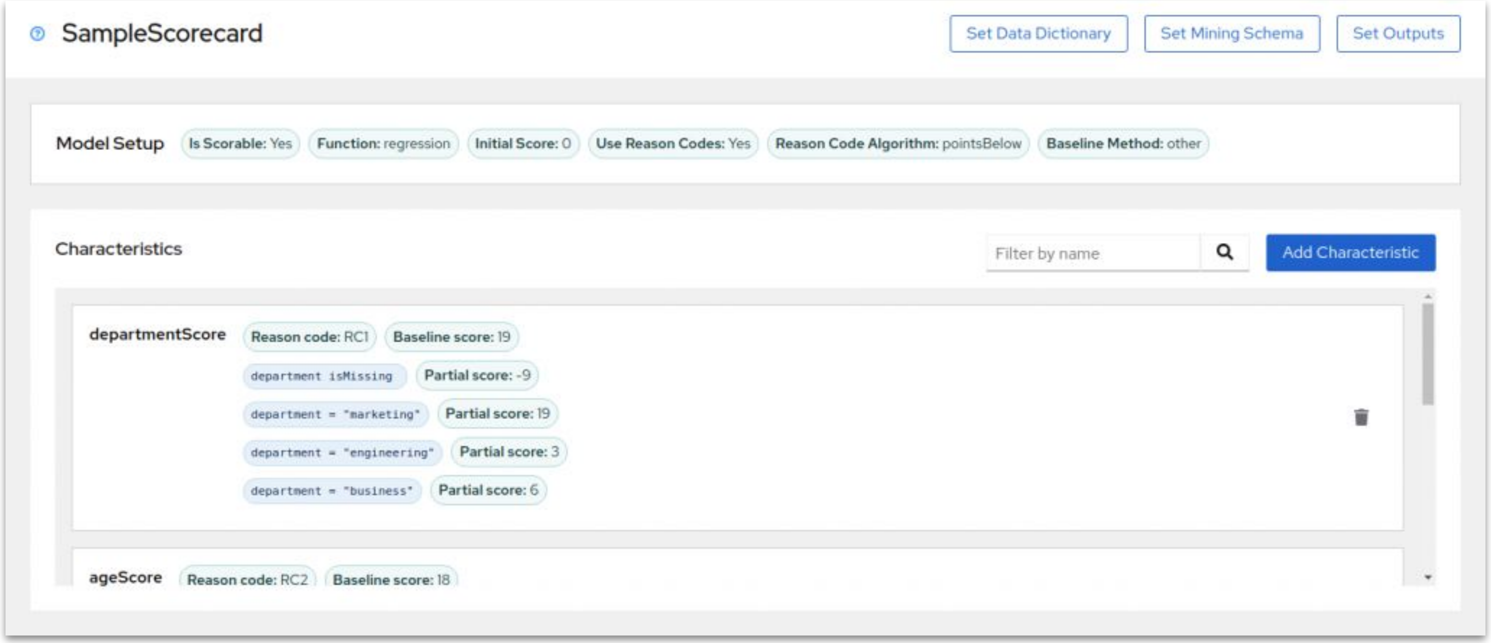
AUTOMATION OF SCORING

Scorecard is a risk management tool used mostly by banks to calculate the risk they take by selling you one of their products.

How risky is a customer?
What are the changes the customer might default on payment?

Based on the customer score we may adapt the product offer - a better interest rate, a higher credit limit, etc.

PMML Scorecard Editor in VS Code



AUTOMATION OF HUMAN-LIKE CONVERSATIONS

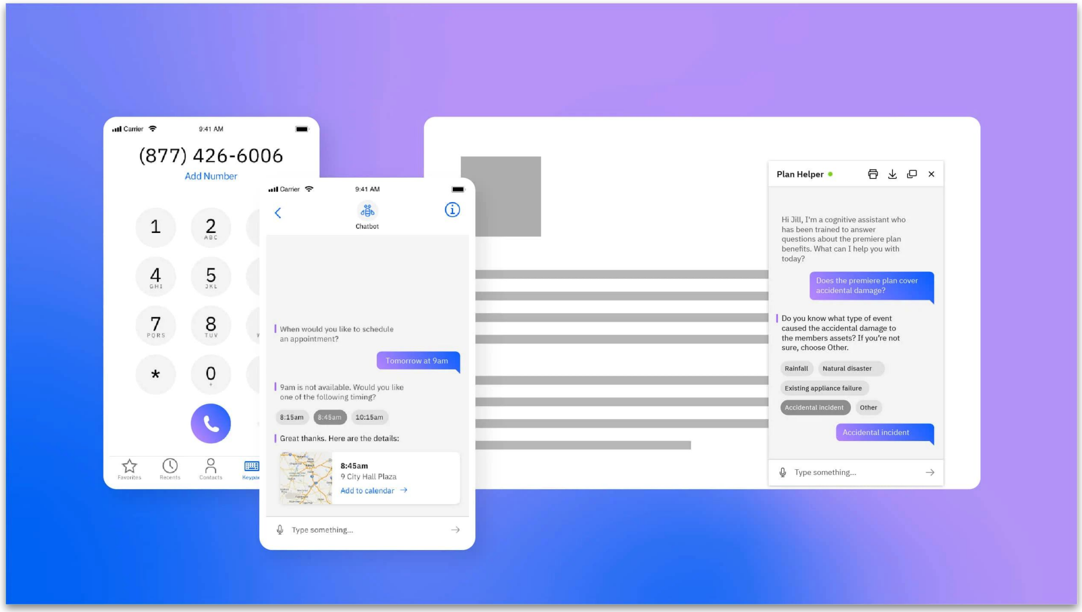
Chatbots

A chatbot is software that simulates human-like conversations with users via text messages on chat or text-to-speech.

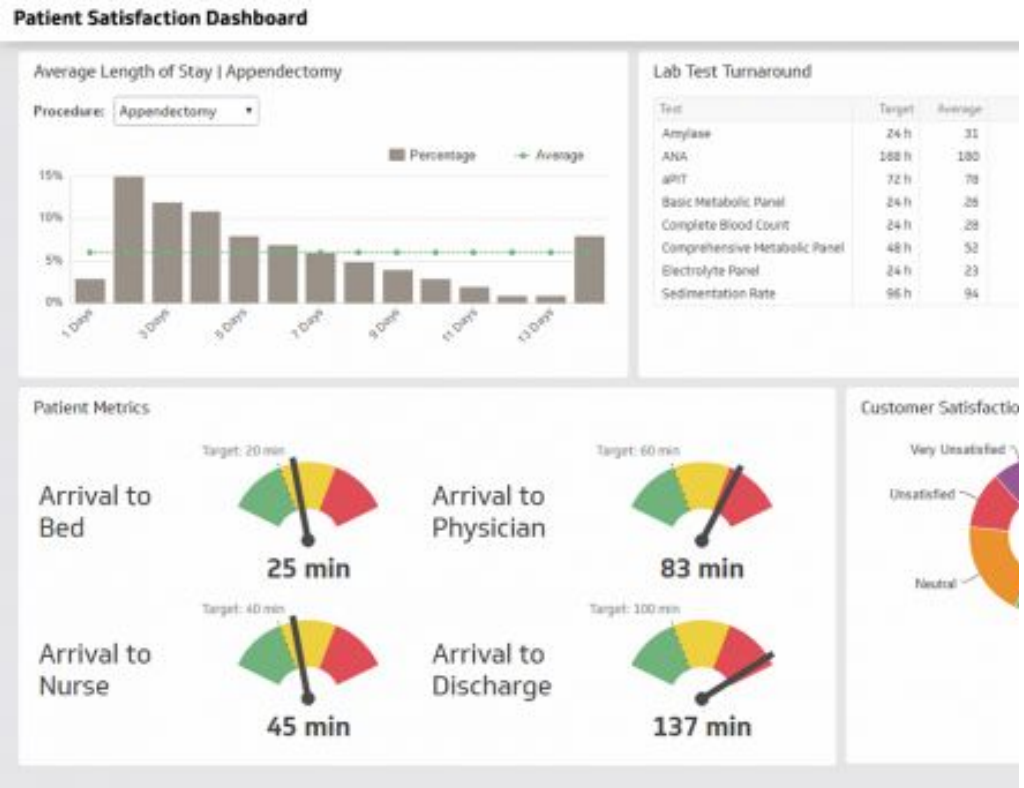
- Customer self-service - call centres, scheduling doctor appointments
- Employee self-service - HR assistants, meeting and scheduling, expenses, people finder

Omnichannel - communication is done on all kinds of channels - phone calls, chat and email on computers, sms messages, ...

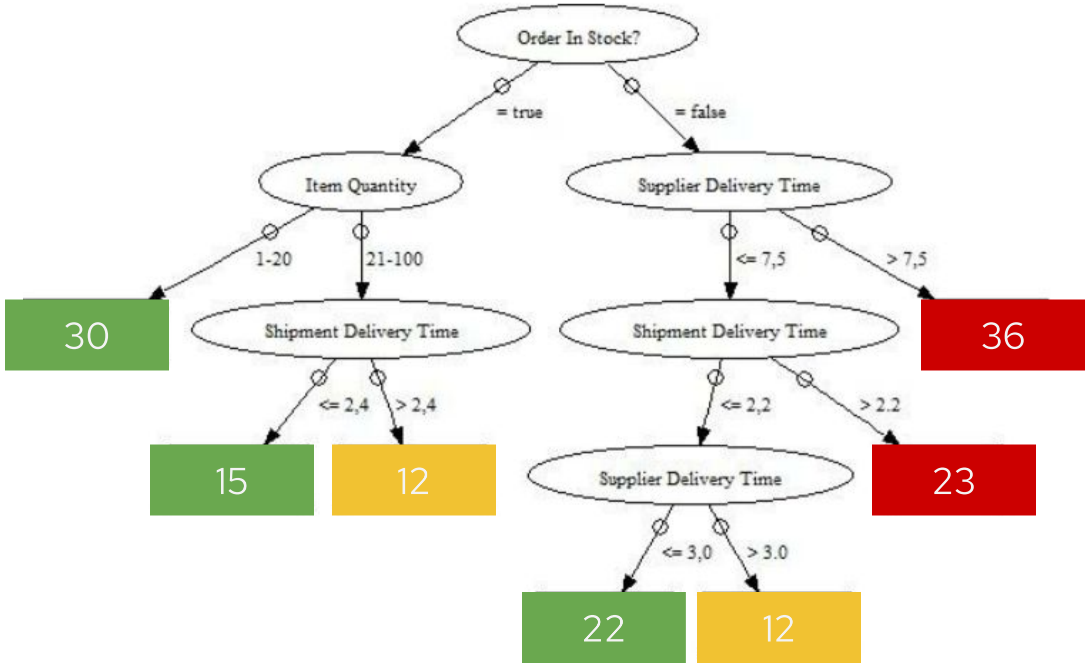
Watson Assistant - COVID-19 response automation



AUTOMATION OF BUSINESS MONITORING



KPI Dependency Tree



Alerts, triggers, ...

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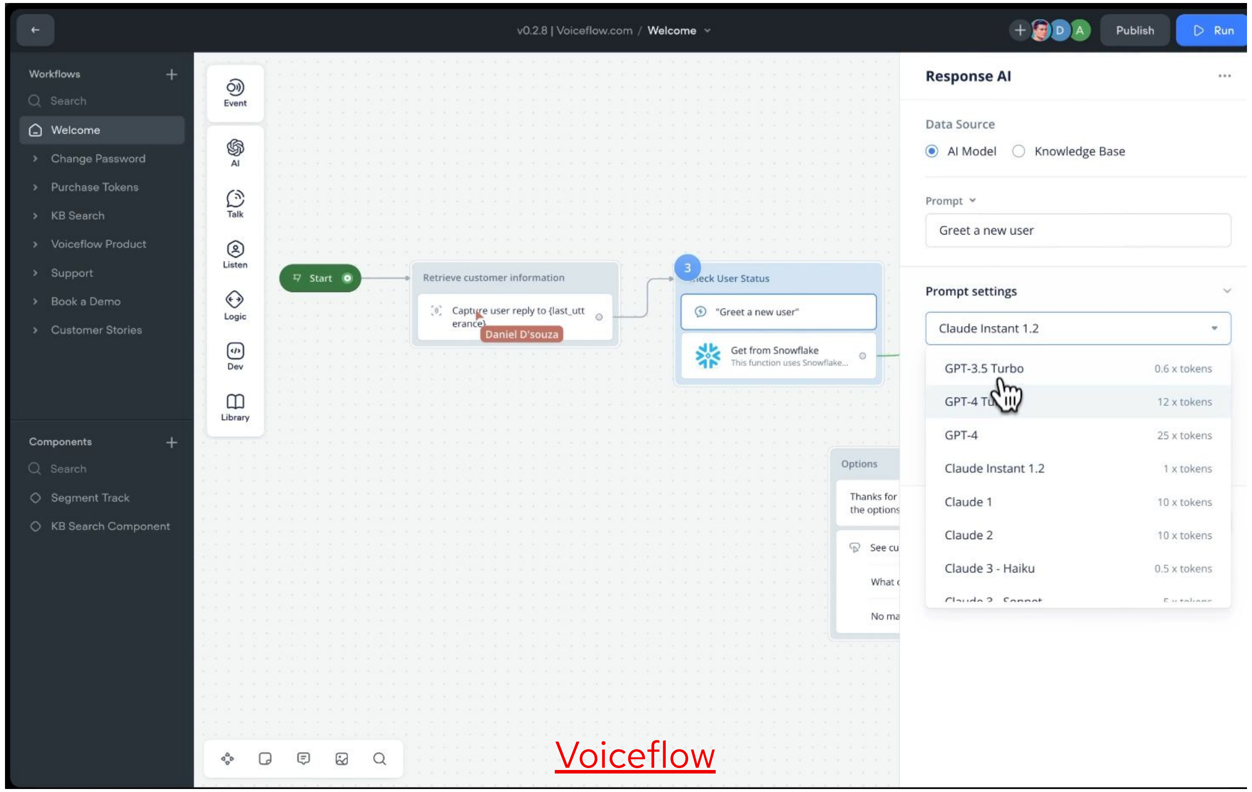
AGENTIC AI

- Agentic AI uses sophisticated reasoning and iterative planning to autonomously solve complex, multi-step problems.
- 4-step process
  - Perceive - gather and process data
  - Reason - using gathered data
  - Act - solve the problem
  - Learn - adapt and be more effective over time
- Originally relied purely on coding
- Nowadays no-code versions are starting to appear
  - “Next-gen workflow engine”



[Voiceflow]

[NVIDIA Blog: What Is Agentic AI?]
[No Code AI Agents]
[Low Code/No Code Platforms]

Choose the right software and services to fit your needs

Thank you, questions?